KENYA MEDICAL TRAINING COLLEGE
FACULTY OF CLINICAL SCIENCES
DEPARTMENT OF CLINICAL MEDICINE

FINAL QUALIFYING EXAMINATION
FOR
DIPLOMA IN CLINICAL MEDICINE & SURGERY

PAPER: SURGERY

DATE: 22ND JUNE, 2022 **TIME: 3 HOURS (9:00AM – 12:00 NOON)**

INSTRUCTIONS

1. This paper consists of:
   - Section 1 (40 Multiple Choice Questions)
   - Section 2 (8 Short Answer Questions)
   - Section 3 (1 Long Answer Question)
2. Attempt **ALL** Questions
3. Write the EXAMINATION **NUMBER** given on all the answer sheets provided and on the question paper.
4. Ensure that all examination answer scripts are handed in at the end of the examination
5. Ensure you sign the examination register provided

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SECTION 1 MULTIPLE CHOICE QUESTIONS (40 MARKS)

1) One of the reasons of recording patients' full name in clerkship is to;
   a) Make a diagnosis
   b) Establish rapport
   c) Assess social economic status
   d) Predilection of diseases
2) Which of the following best describes a surgical procedure performed to aid or confirm diagnosis?
   a) Relantation
   b) Extirpation
   c) Exploratory surgery
   d) Transplant surgery
3) Which of the following refers to surgical fusion of a joint?
   a) Arthrodesis
   b) Arthroplasty
   c) Arthroscopy
   d) Arthrography
4) Fixation of a testis after untwisting of testicular torsion is referred to as;
   a) Orchidectomy
   b) Orchideplasty
   c) Orchidopexy
   d) Ochiectomy
5) In the 'Ebb and Flow'model, ebb phase is characterized by;
   a) Hypovolaemia
   b) Increased basal metabolic rate
   c) Leucocystosis
   d) Increased cardiac output
6) Metabolic stress response to surgery can be reduced by;
   a) Administering steroids
   b) Enhancing bed rest
   c) Adequate analgesic
   d) Maximizing pre-operative fasting.

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7) Pre-operative sedation in an anxious patient can be achieved by administering
   a) Atopine
   b) Cimetidine
   c) Glycopyrronium
   d) Benzodiazepines

8) Secondary post-operative haemorrhage is caused by;
   a) Fall in blood pressure
   b) Infection
   c) Vessel cut during surgery
   d) Prolonged bed rest

9) Concerning would dehiscence;
   a) Occurs from 1st to 4th post-operative day.
   b) Steroids have a role in its management
   c) Is associated with skin colour
   d) Post-surgical haematoma is a risk factor

10) General anaesthetic triad comprises which of the following?
   a) Pain relief, amnesia, muscle relaxation
   b) Unconsciousness, pain relief and muscle relaxation
   c) Amnesia, unconsciousness and muscle relaxation
   d) Immobility, amnesia, unconsciousness

11) Which of the following local anaesthetic drug is an amide?
   a) Lidocaine
   b) Cocaine
   c) Tetracaine
   d) Benzocaine

12) Low grade fever within the first 24 hours after surgery is caused by;
   a) Wound infection
   b) Release of inflammatory medications
   c) Cannula infection
   d) Deep venous thrombosis

13) Which of the following is true concerning a diabetic patient scheduled for an elective surgery?
   a) Should be admitted one day prior to surgery
   b) Fasting period should be minimized
   c) Insulin is contra indicated
   d) Oral hypoglycaemic drugs should be continued up to the day of operation.

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14) Fracture mid shaft left femur in a 13-year-old male patient is managed by;
   a) Internal fixation
   b) Skeletal traction
   c) Application of PoP hip spica
   d) Balanced skin traction

15) Pain that presents most of the times and requires continuous treatment is graded as
   a) Grade II
   b) Grade IV
   c) Grade I
   d) Grade III

16) Avascular necrosis is likely to occur in the following fracture;
   a) Monteggia fracture
   b) Bennett's fracture
   c) Fracture scaphoid
   d) Lateral molleollar fracture

17) In acute osteomyelitis;
   a) X-ray features are evident in the first three days of infections
   b) Commonest cause is streptococcus pyogens
   c) Heals spontaneously
   d) Antibiotics are administered for more than two weeks

18) Which of the following is a feature of sun burst spiculation on an x-ray?
   a) Chronic osteomyelitis
   b) Osteogenic sarcoma
   c) Multiple myeloma
   d) Ewings' sarcoma

19) In management of open fractures;
   a) Surgical toilet and debridement
   b) Antibiotics are started after debridement
   c) Wound cover in Gustillo IIIB is done using local tissues
   d) External fixation is preferred for fracture stabilization.

20) A 24-year-old male patient presents in casualty with pain and inability to use his left shoulder joint following a fall on outstretched hand.
   a) Request for an x-ray
   b) Perform physical examination
   c) Refer to an orthopedic surgeon
   d) Admit in surgical ward

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21) An x-ray of a 75 year old female patient with history of severe pain and stiffness of her right knee joint reveals reduced joint space, multiple osteophytes and genu valgum deformity.

What is the diagnosis?

a) Tuberculosis arthritis
b) Gouty arthritis
c) Pyogenic arthritis
d) Osteoarthritis

22) Epiphyseal (Salter Harris) injury with fracture of the articular surface with extension across the growth plate into metaphysisis is classified as;

a) Type IV
b) Type V
c) Type III
d) Type II

23) The following test is performed to detect tear of anterior cruciate ligament:

a) Pivot shift test
b) Posterior drawers test
c) Barrows test
d) Patella tap

24) Which one of the following is an x-ray feature of rickets?

a) Cortical thickening
b) Narrowing and cupping of metaphysis
c) Narrowing of the growth plate
d) Widening and fanning of metaphysis

25) The following is a developmental vascular disease that affects proximal femoral epiphysis:

a) Slipped upper femoral epiphysis
b) Perthe's disease
c) Osteogenesis imperfect
d) Congenital dislocation of the hip joint

26) Typical deformities of the foot in congenital club foot;

a) Eversion, adduction, and plantar flexion
b) Abduction, adduction, and plantar flexion
c) Inversion, adduction and dorsi-flexion
d) Inversion, adduction and plantar flexion

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27) Which one of the following is an indication for emergency laparotomy?
   a) Haematemesis in peptic ulcer disease
   b) Cholelithiasis
   c) Pancreatitis
   d) Diaphragmatic rupture

28) Which of the following is a 'mural' cause of intestinal obstruction?
   a) Bezoar
   b) Cystic fibrosis
   c) Volvulus
   d) Adhesions

29) Which of the following is true about small bowel obstruction?
   a) Present with lower abdominal colic
   b) Is commonly of a sudden onset
   c) Vomiting is a late presentation
   d) Barium enema is the preferred investigative modality

30) Concerning post hepatic (obstructive) jaundice;
   a) Urine contains urobilinogens
   b) Liver enzymes are within normal values
   c) Conjugated serum bilirubin are elevated
   d) Alkaline phosphatase levels are reduced

31) The following is a clinical feature of superficial second degree burns;
   a) Waxy white appearance
   b) Blisters
   c) Absent capillary return on pressure
   d) Leathery texture

32) A chronic facial ulcer with raised edges is most likely;
   a) Tuberculous ulcer
   b) Rodent's ulcer
   c) Syphilitic ulcer
   d) Healing ulcer

33) One of the following is true about carotid body tumor (chemodectoma):
   a) Fine needle aspirate cytology is contraindicated
   b) Commonly occurs in teenagers
   c) Surgery is contraindicated
   d) Is a differential diagnosis of posterior neck swelling

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34) A head injury patient who was fully alert and oriented developed rapid neurological deterioration with contralateral hemiparesis, reduced consciousness, and ipsilateral pupillary dilatation one hour after admission.

What is the most likely diagnosis?

a) Acute subdural haematoma
b) Acute epidural haematoma
c) Subarachnoid haemorrhage
d) Intraventricular haemorrhage

35) Primary brain injury can occur due to;

a) Hypoxia
b) Hypotension
c) Meningitis
d) Penetrating injury

36) The following scrotal swelling arises from subcutaneous tissue;

a) Cellulitis
b) Sebaceous cysts
c) Haematocele
d) Epidermoid cancer

37) Which one of the following is an irritative symptom in a patient with benign prostatic hypertrophy?

a) Hesitancy
b) Dribbling
c) Urge incontinence
d) Intermittent stream

38) Concerning voluntary medical male circumcision (VMMC);

a) Dorsal block is achieved by injecting lignocaine at 12 and 6 o'clock sites
b) Scrubbing is unnecessary
c) Hypospadias is an indication
d) Sexual intercourse should be avoided for six weeks after the procedure

39) One of the following is true about direct inguinal hernia:

a) Commonly strangulates
b) Reduces spontaneously on lying supine
c) Can be controlled by applying pressure over internal inguinal ring
d) Recurrence after surgery is uncommon.

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40) Fracture of the base of first metacarpal extending to involve the carpo-metacarpal joint is referred to as;

a) Smith fracture
b) Pilon fracture
c) Galleazi fracture
d) Bennett's fracture

SECTION 2 SHORT ANSWER QUESTIONS (40 MARKS)

41) A 40-year-old was brought to casualty after having suffered blunt chest injury. After history taking and physical examination, a diagnosis of left haemothorax was made. Enumerate five (5) clinical features that support this diagnosis. **(5 marks)**

42) A 25-year-old female patient weighing 55kg sustained 2nd degree burns involving the entire anterior trunk, perineum as well as entire right and left anterior thighs. Using Parkland's formula, calculate the amount of I/V fluids to be administered in the first 24 hours. **(5 marks)**

43) A middle aged male patient presents to you with an ulcer on his right leg. Outline five (5) enquiries you would consider to make in history taking. **(5 marks)**

44) List five (5) indications of limb amputation. **(5 marks)**

45) Outline the management of penetrating joint injury. **(5 marks)**

46) Outline the principles of treating chronic osteomyelitis. **(5 marks)**

47) List five (5) complications of Colle's fracture. **(5 marks)**

48) Describe the ortolani test of detecting congenital dislocation of the hip joint. **(5 marks)**

SECTION 3 LONG ANSWER QUESTION (20 MARKS)

49) A 32 year old male patient has been admitted in surgical ward with a diagnosis of intestinal obstruction.

a) List four (4) GIT clinical features that support this diagnosis. **(4 marks)**

b) Outline the pre-operative management of this patient **(12 marks)**

c) List four (4) complications of intestinal obstruction. **(4 marks)**

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